

GUJARAT TECHNOLOGICAL UNIVERSITY
BE –SEMESTER 1&2(NEW SYLLABUS)EXAMINATION- WINTER 2018

Subject Code: 3110018**Date: 04-01-2019****Subject Name: Physics****Time: 10:30 am to 01:00 pm****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

- Q.1** (a) Give assumptions of classical free electron theory. **03**
(b) Explain mechanism of superconductivity. **04**
(c) What is photovoltaic effect. Explain construction and working of solar cell. **07**

- Q.2** (a) Give difference between N type and P type semiconductors. **03**
(b) Derive an expression for joint density of states. **04**
(c) Explain Kronig Penney model in detail. **07**

OR

- (c) Explain properties of superconductors. **07**
Q.3 (a) What are hot probe method. **03**
(b) Explain fermi levels. **04**
(c) Explain classification of materials as conductors, insulators and semiconductors. **07**

OR

- Q.3** (a) Give difference between intrinsic and extrinsic semiconductors. **03**
(b) Explain drift and diffusion current. **04**
(c) Explain direct and indirect band gap with E-k diagrams. **07**
Q.4 (a) Define superconductivity and critical temperature. **03**
(b) Discuss fermi golden rule. **04**
(c) Explain diffusion mechanism in detail. **07**

OR

- Q.4** (a) Define radiative and non-radiative transitions. **03**
(b) Explain emission and absorption. **04**
(c) Explain experimental procedure for DLTS. **07**
Q.5 (a) The critical temperature of Nb is 9.15 K. At zero kelvin, the critical field is 0.196 T. Calculate the critical field at 6 K. **03**
(b) Explain Drude model. **04**
(c) Why two probe method for resistivity measurement failed and hence explain four probe method. **07**

OR

- Q.5** (a) Give success and drawback of classical free electron theory. **03**
(b) Derive expression of electron concentration in conduction band. **04**
(c) Discuss UV – VIS method for band gap measurement of semiconductors. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-I &II (NEW) EXAMINATION – SUMMER-2019****Subject Code: 3110018****Date: 03/06/2019****Subject Name: Physics****Time: 10:30 AM TO 01:00 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

		Marks
Q.1	(a) Give formation and applications of SQUID.	03
	(b) Explain intrinsic and extrinsic semiconductors with necessary diagram.	04
	(c) What is PN junction diode? What is external bias? Describe its forward and reverse bias conditions with appropriate diagram.	07
Q.2	(a) What is photo conductivity, photoluminescence, phototransistor?	03
	(b) Calculate the energy gap of Si, given that radiation of wavelength 11,000 Å is incident on it. Also find allowed wavelength for Ge with energy gap 0.90 eV.	04
	(c) Write a note on energy band diagram and formation of energy bands.	07
	OR	
	(c) Define Hall effect and Hall coefficient. Derive equation to find Hall voltage. What does it signify?	07
Q.3	(a) Differentiate between soft and hard superconductors	03
	(b) What is London penetration depth? Derive its equations.	04
	(c) Derive equations for n-type semiconductor to determine dependence of fermi level on temperature and doping concentration.	07
	OR	
Q.3	(a) The critical current density equal to $1.71 \times 10^8 \text{ A/m}^2$ is required to change a superconducting wire of radius 0.5 mm at 4.2 K. If the critical temperature of the material is 7.18 K, calculate the maximum value of the critical magnetic field.	03
	(b) Explain BCS theory for superconductivity.	04
	(c) Write a note on metal semiconductor junctions.	07
Q.4	(a) Write a note on exciton.	03
	(b) Give details of applications of solar cell (at least 4)	04
	(c) What is radiative and non-radiative transition. Explain in brief the optical joint density of states.	07
	OR	
Q.4	(a) What are direct and indirect band gap?	03
	(b) What is deep level transient spectroscopy (DLTS)? Give its experimental procedure.	04
	(c) Discuss the technique to obtain band gap by UV-Vis spectroscopy using absorption or transmission.	07
Q.5	(a) What are capacitance voltage measurements?	03
	(b) Consider n-type silicon semiconductor with a length of 100 μm , cross sectional area 10^{-7} cm^2 , minority charge carrier life time 10^{-6} s , μ_e is 0.13 m^2/Vs and μ_h is 0.05 m^2/Vs . Find (a) Electron transit time	04

- (b) Photo conductor gain when voltage applied to the photoconductor is 12 V
- (c) Discuss Van Der Pauw method. **07**
- OR**
- Q.5** (a) What is the cause and remedy for optical loss in photovoltaic cell? **03**
- (b) State principle and discuss working of semiconductor laser. **04**
- (c) What is photovoltaic effect? Explain construction and working of a solar cell with suitable diagram **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER– I & II (NEW) EXAMINATION – WINTER 2019****Subject Code: 3110018****Date: 02/01/2020****Subject Name: Physics****Time: 10:30 AM TO 01:00 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

Marks

- | | | |
|------------|---|-----------|
| Q.1 | (a) Enlist the assumptions of free electron theory. | 03 |
| | (b) Give the difference between Direct and Indirect band gap. | 04 |
| | (c) Explain forward and reverse bias conditions in PN junction diode. | 07 |
| Q.2 | (a) Define Intrinsic and extrinsic semiconductor . | 03 |
| | (b) The thermal and electrical conductivity of Cu at 20°C are $390 \text{ Wm}^{-1}\text{K}^{-1}$ and $5.87 \times 10^7 (\Omega\text{m})^{-1}$ respectively. Calculate the Lorentz number. | 04 |
| | (c) Explain Schottky diode in detail. | 07 |
| | OR | |
| | (c) Explain the dependence of Fermi level on temperature | 07 |
| Q.3 | (a) Explain Drude model | 03 |
| | (b) Fermi energy of a given substance is 7.9 eV. What is the average energy and speed of electron in this substance at 0 K? | 04 |
| | (c) Explain photovoltaic effect. With required diagrams discuss construction and working of solar cell. | 07 |
| | OR | |
| Q.3 | (a) Write a short note on exciton. | 03 |
| | (b) Consider two-dimensional square lattice of side $3.0 \text{ \AA}$. At what electron momentum values do the sides of first Brillouin zone appear? What is the energy of free electron with this momentum? | 04 |
| | (c) Derive an equation of joint density of states. | 07 |
| Q.4 | (a) Define Hall effect. Give its physical significance. | 03 |
| | (b) 2.0 cm wide and 1.0 mm thick copper strip is placed in a magnetic field 1.5 Wb/m^2 perpendicular to the strip. Suppose a current of 200 A is set up in the strip what will be the Hall potential appeared across the strip? ($n = 8.4 \times 10^{28} \text{ electrons/m}^3$). | 04 |
| | (c) Discuss UV-VIS method for band gap measurement of semiconductor. | 07 |
| | OR | |
| Q.4 | (a) Discuss Fermi golden rule. | 03 |
| | (b) The transmitted intensity is 0.4 times intensity of incident light. If this light is incident on a semiconductor having a thickness of 0.5 cm then find absorption coefficient. | 04 |
| | (c) Explain four probe method. Derive an equation to calculate resistivity of a thin sample. | 07 |

- Q.5** (a) Write short note cryotron. **03**
(b) Explain London's penetration depth. **04**
(c) Give the difference between type 1 and type 2 superconductor. **07**

OR

- Q.5** (a) Write short note on SQUID. **03**
(b) Calculate the critical current for a superconducting wire of lead having a diameter of 2 mm at 2 K. Critical temperature for lead is 4 K and $H_c(0) = 6.5 \times 10^4$ A/m. **04**
(c) Explain the properties of superconductors in detail. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-I & II (NEW) EXAMINATION – WINTER 2020****Subject Code:3110018****Date:17/03/2021****Subject Name:Physics****Time:10:30 AM TO 12:30 PM****Total Marks:56****Instructions:**

1. Attempt any FOUR questions out of EIGHT questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

		Marks
Q.1	(a) Explain how the materials are classified into conductors, semiconductors and insulators on the basis of energy band diagram.	03
	(b) Define Fermi level for intrinsic semiconductor. How it will change in p-type and n-type semiconductors. Explain with suitable diagrams.	04
	(c) Explain formation and working of p-n junction diode in forward and reverse biasing by proper diagrams and its I-V characteristics.	07
Q.2	(a) Explain absorption, spontaneous emission and stimulated emission in semiconductors with proper diagrams.	03
	(b) Write down an expression for the probability of occupancy of a particular energy state of an electron in an intrinsic semiconductor. Explain it from the graph at 0° K and at room temperature	04
	(c) Describe formation of Energy bands by using Kronig-Penney Model.	07
Q.3	(a) Explain n-type and p-type semiconductors with suitable diagrams	03
	(b) Explain direct band gap and indirect band gap semiconductors with proper E-k diagrams.	04
	(c) Derive an expression for density of energy states in metals.	07
Q.4	(a) Write down any three differences between intrinsic and extrinsic semiconductors.	03
	(b) Explain construction and working of Schottky junction	04
	(c) Give the names of semiconductor optoelectronic devices, their characteristics and applications.	07
Q.5	(a) Explain law of mass action.	03
	(b) Explain Photovoltaic Effect and derive an expression for photo voltage.	04
	(c) Write down various properties of superconductors.	07
Q.6	(a) Explain (i) Non-radiative transitions and (ii) Exciton .	03
	(b) Explain construction and working of a solar cell by proper diagrams and I-V characteristics.	04
	(c) Explain UV-VIS spectroscopy and how to find energy band gap of a material from this technique.	07
Q.7	(a) At 0 magnetic field, a superconducting Tin has a critical temperature of 3.7 K. At 0 K, critical magnetic field is 0.306 T. Calculate the critical magnetic field at 2 K	03
	(b) Write down the applications of superconductors.	04
	(c) Derive a formula for carrier concentration in n-type semiconductor	07

- Q.8**
- (a) Describe Hall Effect with a suitable diagram. **03**
 - (b) A semiconductor has Hall coefficient $3.75 \times 10^{-4} \text{ m}^3/\text{C}$. The resistivity of the sample is $7.21 \times 10^{-3} \text{ ohm m}$. Calculate the mobility and density of charge carriers. Given that the magnitude of charge on electron = $1.6 \times 10^{-19} \text{ C}$. **04**
 - (c) Explain **07**
 - (a) Type I and Type II superconductors 02
 - (b) Low T_c and High T_c superconductors 02
 - (c) SQUIDS and its applications 03

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-1/2 EXAMINATION – WINTER 2021****Subject Code:3110018****Date:22/03/2022****Subject Name:Physics****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Define absorption, spontaneous emission and stimulated emission for bulk semiconductors.	03
	(b) Write a short note on P-N junction diode.	04
	(c) Give assumptions of classical free electron theory and discuss its limitations.	07
Q.2	(a) Find the temperature at which there is 1% probability that a state with energy 2eV is occupied. Given that Fermi energy is 1.5 eV.	03
	(b) What is drift and diffusion current in p-n junction diode?	04
	(c) Explain the Kronig-Penny model of solids and show that how it explains the origin of band gap in solids.	07
	OR	
	(c) Explain intrinsic and extrinsic (N & P type) semiconductors with the help of energy band diagram.	07
Q.3	(a) In an N-type semiconductor, the Fermi level lies 0.3 eV below the conduction band at room temperature. If the temperature is increased to 330°K, Find the position of Fermi level.	03
	(b) Write an expression for Fermi Dirac distribution function $f(E)$. Show that at all temperatures ($T > 0K$) probability of occupancy of Fermi level is 50%.	04
	(c) Discuss the effect of temperature on the Fermi level in extrinsic (N & P type) semiconductors.	07
	OR	
Q.3	(a) Find the concentration of holes and electrons in N-type silicon if the conductivity is $0.1 \Omega\text{-cm}^{-1}$, mobility of electrons is $1300 \text{ cm}^2/\text{V-s}$ and total carrier concentration is $1.5 \times 10^{10} \text{ carriers / cm}^3$.	03
	(b) Establish the relation between Einstein's coefficients.	04
	(c) Explain Meissener's effect in detail and show that for superconductor, $\chi_m = -1$.	07
Q.4	(a) Write a short note on effective mass of electron.	03
	(b) What is mass action law? Explain Schottky junction.	04
	(c) Explain Drude model and discuss how it is used for Hall measurements and magnetoresistance.	07
	OR	
Q.4	(a) What is an exciton? What is DLTS? Define Hall mobility.	03

- (b) What is Fermi level and Fermi energy? **04**
What is Photovoltaic Cell?
- (c) Explain four point probe method with diagram for the measurement of resistivity of bulk sample. **07**
- Q.5** (a) Explain Fermi Golden rule for transition probability. **03**
(b) What is Josephson junction? Write a short note on SQUID. **04**
(c) Explain how to measure band gap of the semiconductor using UV-Vis spectroscopy. **07**
- OR**
- Q.5** (a) Calculate the critical current for a superconducting wire of lead having a diameter of 1mm at 4.2 K. Critical temperature for lead is 7.18 K and $H_c(0) = 6.5 \times 10^4$ A/m. **03**
(b) Write a short note on Hot-point probe measurement technique. **04**
(c) What is superconductivity? Explain any six properties of superconductor. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-I & II(NEW)EXAMINATION – SUMMER 2022****Subject Code:3110018****Date:04-08-2022****Subject Name:Physics****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) What is physical significance of the negative effective mass?	03
	(b) Find the probability of an electron occupying an energy level 0.02 eV above the Fermi level at 200 K and 400 K in a material.	04
	(c) Derive an expression for electrical and thermal conductivity in a material and hence verify Wiedemann-Franz law.	07
Q.2	(a) Show that for an intrinsic semiconductor the Fermi level lies at the middle of the energy gap.	03
	(b) The intrinsic carrier density of given semiconductor is $1.5 \times 10^{16} \text{ m}^{-3}$. If the mobility of the electrons and holes are 0.13 and $0.05 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$ respectively, calculate the conductivity.	04
	(c) Obtain an expression for concentration of holes in valance band in intrinsic semiconductor.	07
	OR	
	(c) Derive the formula for concentration of electrons in conduction band in n-type semiconductor.	07
Q.3	(a) Write the difference between spontaneous emission and stimulated emission.	03
	(b) Explain Ohmic junction with necessary diagram.	04
	(c) What do you mean by joint density of state? Derive mathematical expression for optical joint density of states.	07
	OR	
Q.3	(a) Write short note on exciton.	03
	(b) If the light having wavelength of $4000 \text{ \AA}$ falls on semiconductor having bandgap of 2.1 eV. Assuming mass of electron in conduction band and valance band is same as rest mass of electron, calculate the optical joint density of states for given semiconductor.	04
	(c) For bulk semiconductor show that the ratio of Einstein's co-efficient is directly proportional to cube of frequency.	07
Q.4	(a) Write the drawbacks of two probe method.	03
	(b) An n-type Ge sample has donor density of 10^{21} m^{-3} . It is arranged in a Hall effect experiment having $B_z = 0.5 \text{ Wb/m}^2$ and $J_x = 500 \text{ A/m}^2$. Find Hall voltage if the width of sample is 3 mm.	04
	(c) What is Hall effect? Obtain expressions for Hall Voltage and Hall mobility.	07

OR

- Q.4** (a) Explain Hot point probe measurement. **03**
(b) A 20.0 mm wide and 1.0 mm thick silver strip is placed in 1.5 Wb/m^2 magnetic field in such a way that magnetic field remains perpendicular to strip. A current of 200 A is set-up in the strip. Calculate the Hall voltage of the strip. (given: $n = 8.4 \times 10^{28} \text{ m}^{-3}$) **04**
(c) Explain Current-Voltage characteristic of Solar cell. **07**

- Q.5** (a) Define: (1) Critical Temperature (2) Critical Magnetic field (3) Critical Current density. **03**
(b) For specimen of V_3Ga , the critical fields are $1.4 \times 10^5 \text{ A/m}$ and $4.2 \times 10^5 \text{ A/m}$ at 14 K and 13 K respectively. Calculate the value of transition temperature. **04**
(c) Write and explain characteristics of superconductors. **07**

OR

- Q.5** (a) The critical temperature for Hg with isotopic mass 199.5 is 4.185 K. Calculate its critical temperature when its isotopic mass changes to 203.4. **03**
(b) Write the difference between Type-I and Type-II superconductor. **04**
(c) Explain BCS theory. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-I & II(NEW) EXAMINATION – WINTER 2022****Subject Code:3110018****Date:04-03-2023****Subject Name:Physics****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) What are the postulates for free electron theory?	03
	(b) Compare Direct Band Gap semiconductors and Indirect Band Gap semiconductors	04
	(c) Explain in detail how Kronig and Penney model explains the origin of band gap in semiconductors.	07
Q.2	(a) Show that Probability of occupancy of Fermi energy level is 0.5 at $T > 0$ K.	03
	(b) What is effective mass of electron? Derive necessary expression for it.	04
	(c) (i) Consider an energy level lying 0.01 eV above Fermi level. What is the probability of this level not being occupied by an electron at 300 K. (ii) At What temperature we can expect a 20% probability the electrons in a metal can have which is lying 1% above its Fermi energy? Fermi energy of the given metal is 5.5 eV.	07
	OR	
	(c) Make use of necessary assumption for intrinsic semiconductors and derive the formula for density of hole in valence band.	07
Q.3	(a) Compare Intrinsic and Extrinsic Semiconductors.	03
	(b) For a given intrinsic semiconductor with $E_g = 2.0$ eV, calculate the position of Fermi Level at 300 K. ($m_e^* = 0.16m_0$ and $m_h^* = 0.32m_0$, m_0 is rest mass of an Electron).	04
	(c) Explain the Einstein's theory about interaction between matter and radiation in detail along with absorption and emission with necessary equation.	07
	OR	
Q.3	(a) What is Luminescence? Discuss its types.	03
	(b) Define :(i) Forbidden Gap (ii)Fermi Function, (iii) Density of states (iv) Drift Current	04
	(c) (i) Find the concentration of holes and electron in (1) n-Type and (2) p-Type semiconductor having conductivity $5 \times 10^4 \text{ } \Omega/\text{m}$ at 300 K. Mobility of electrons and holes is $1500 \times 10^{-4} \text{ m}^2/\text{vSec}$ and $500 \times 10^{-4} \text{ m}^2/\text{vSec}$ respectively and $n_i = 9.5 \times 10^{16} \text{ m}^{-3}$. (ii) A Light Emitting Diode has a band gap of 1.96 eV. Calculate the wavelength of the emitted radiation	07

- Q.4 (a)** What is hot point probe method? Explain **03**
- (b)** What are Excitons? Discuss the types of Exciton. **04**
- (c)** Illustrate Four probe method for thin sheet and bulk sample in detail. **07**
- OR**
- Q.4 (a)** What do you mean by Photo Voltaic Effect? How it is differ from Photo Electric effect? **03**
- (b)** Explain Optical Gain and Loss in Solar Cell. **04**
- (c)** (i) The critical field for a superconducting sample is 10^6 Am^{-1} at 9.58 K and $2 \times 10^6 \text{ Am}^{-1}$ at 0 K. Determine the T_c value. **07**
- (ii) Calculate the critical current through a long thin superconducting wire of diameter 0.5 mm. The critical field is 9.2 kA/m.
- Q.5 (a)** Define following terms. **03**
- Magnetic Levitation, Persistent Current, Isotope effect
- (b)** Compare Type-I and Type-II Superconductivity. **04**
- (c)** Explain UV-Vis spectroscopy method for band gap measurement of semiconductors. **07**
- OR**
- Q.5 (a)** Show that superconductors are perfect diamagnet. **03**
- (b)** Identify three important factors that defines a superconducting state in detail. **04**
- (c)** (i) Explain how the materials are classified into conductors, semiconductors and insulators with the help of energy band diagrams. **07**
- (ii) What are the applications of Hall effect measurements?

Seat No.: _____

Enrolment No. _____

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER-I & II(NEW) EXAMINATION – SUMMER 2023

Subject Code:3110018

Date:01-08-2023

Subject Name:Physics

Time:10:30 AM TO 01:00 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

MARKS

Q.1	(a) Define: Recombination and carrier generation	03
	(b) Differentiate: Spontaneous and Stimulated emission.	04
	(c) Explain Energy Band Diagrams in solids with the necessary diagram.	07
Q.2	(a) State assumption of classical free electron gas theory.	03
	(b) Differentiate: Direct and Indirect bandgap in semiconductors	04
	(c) Explain: Working of the p-n junction diode and state its applications.	07
	OR	
	(c) Write a short note on DOS (Density of State)	07
Q.3	(a) State: Fermi function	03
	(b) Differentiate: Intrinsic and Extrinsic Semiconductor.	04
	(c) Explain the working of solar cells with the necessary diagram.	07
	OR	
Q.3	(a) Define: Ohmic junction	03
	(b) Explain: Exciton	04
	(c) Write a brief note on Kronig-Penney Model.	07
Q.4	(a) Define: Superconductivity and Transition temperature T_c	03
	(b) Write down applications associated with SQUID.	04
	(c) Explain the Meissner effect and prove that superconductors are perfect diamagnetic.	07
	OR	
Q.4	(a) Define: Absorption of Light	03
	(b) Explain: Hot Point probe measurements.	04
	(c) Write a brief note on the density of electrons in the conduction band.	07
Q.5	(a) What do you mean by drift and diffusion current?	03
	(b) Differentiate: Type I and Type II superconductors.	04
	(c) Explain: Four Point Probe technique (for Bulk sample)	07

OR

- Q.5**
- | | | |
|------------|---|-----------|
| (a) | State: Formula for Fermi's Golden Rules. | 03 |
| (b) | Calculate the critical current which can flow through a long thin superconducting wire of diameter 10^{-3} m. Given $H_c = 7.9 \times 10^3$ amp/m | 04 |
| (c) | Write a brief note on Visible Spectroscopy and state its applications. | 07 |

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-I (NEW) EXAMINATION – WINTER 2023****Subject Code:3110018****Date:12-01-2024****Subject Name:Physics****Time:02:30 PM TO 05:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

- Q.1** (a) Mention any three postulates of classical free electron theory proposed by Lorents and Drude. **03**
- (b) (i) State any two points of differences between intrinsic and extrinsic semiconductors. **04**
- (ii) State any two points of differences between n -type and p -type semiconductors.
- (c) What is superconductivity? Discuss any three properties of superconductors in detail. **07**

- Q.2** (a) Obtain Fermi factor $f(E)$ for energy 0.01 eV above Fermi energy (E_f) at 500 K . **03**
- (b) State and discuss Wiedemann-Franz law. **04**
- (c) Define density of states and derive its expression. **07**

OR

- (c) What is E-k diagram? Discuss direct and indirect band gap semiconductors based on E-k diagram. **07**

- Q.3** (a) (i) Write fermi distribution function and mention physical significance of it. **03**
- (ii) Mention any two applications of solar cell.
- (b) Derive formula of drift and diffusion current densities. **04**
- (c) (i) Draw characteristic curve for p - n junction diode. **07**
- (ii) Discuss forward bias characteristics with proper circuit diagram.
- (iii) Discuss reverse bias characteristics with proper circuit diagram.

OR

- Q.3** (a) A potential barrier of 0.6 V exists across of p - n junction. The intensity of the electric field in this region is $2.4 \times 10^6\text{ V/m}$. Find the width of depletion region. **03**
- (b) (i) Mention any two applications of LED . **04**
- (ii) Calculate the value of band gap for a LED for the emission of blue light of wavelength 452 nm . ($h = 6.63 \times 10^{-34}\text{ J s}$, $c = 3 \times 10^8\text{ m/s}$)
- (c) Derive formula of fermi energy level for n -type semiconductor. **07**

- Q.4** (a) (i) What is called optoelectronic device? **03**
- (ii) Name the types of metal-semiconductor junction.

- (iii) What is called exciton? 04
- (b) Discuss Frenkel and Wannier-Mott exciton in brief. 04
- (c) What is photovoltaic effect? Explain construction and working of solar cell in detail. 07

OR

- Q.4** (a) (i) What is Hall effect? 03
- (ii) The Hall coefficient of a semiconducting material is $2.4 \times 10^{-4} \text{ m}^3/\text{C}$. Calculate the mobility and carrier concentration of the carrier if resistivity is $6 \times 10^{-3} \Omega \text{ m}$.
- (b) Discuss Fermi's golden rule for transition probability. 04
- (c) Discuss in detail UV-VIS method for band gap measurement of semiconductor. 07

- Q.5** (a) Calculate the critical current through a long thin superconducting wire of radius 10^{-3} m . The critical magnetic field is $6.2 \times 10^3 \text{ A/m}$. 03
- (b) State any four points of differences between spontaneous and stimulated emission. 04
- (c) Discuss in detail experimental procedure for DLTS. 07

OR

- Q.5** (a) The critical temperature of a superconductor is 7.5 K. Calculate the critical field at 5 K. At 0 K the critical field is 0.24 T. 03
- (b) Write a short note on London penetration depth. 04
- (c) (i) Discuss BCS theory of superconductivity in brief. 07
- (ii) Draw constructive diagram of SQUID and mention its application.

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER-I & II (NEW) EXAMINATION – SUMMER 2024

Subject Code:3110018

Date:10-07-2024

Subject Name:Physics

Time:02:30 PM TO 05:00 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

	Marks
Q.1 (a) Define following terms. i. Critical Temperature ii. Persistent Current iii. Isotope effect	03
(b) Explain hot point probe method with necessary diagrams.	04
(c) Explain in detail how Kronig and Penney model explains the origin of band gap in semiconductors.	07
Q.2 (a) Compare n-Type and p-Type Semiconductors.	03
(b) Distinguish between Type-I and Type-II Superconductivity.	04
(c) Determine the necessary formula for density of hole in Valence Band for Intrinsic Semiconductors.	07
OR	
(c) What are Einstein's coefficients? Give relation between them and discuss the result.	07
Q.3 (a) Define: i. Energy Gap ii. Fermi Function iii. Density of states	03
(b) Estimate the position of Fermi Level at 300 K for a material with $E_C = 0.8$ eV and $E_V = 0.5$ eV ($m_e^* = 0.14m_0$ and $m_h^* = 0.28m_0$, m_0 is rest mass of an electron).	04
(c) Illustrate Four probe method for thin sheet and bulk sample in detail.	07
OR	
Q.3 (a) What are the postulates for free electron theory?	03
(b) In an intrinsic semiconductor, energy gap is 1.3 eV. Evaluate the ratio between its conductivity at temperature 700 K and 400 K.	04
(c) Show that Junction capacitance is inversely proportional to square root of Reverse biasing Voltage.	07
Q.4 (a) What do you mean by Photo Voltaic Effect? Write its three applications in the field of engineering.	03
(b) Using Fermi function, evaluate the temperature at which there is 20 % probability that an electron in solid will have energy 0.3 eV above E_F of 3 eV.	04

- (c) Explain at least seven properties of Superconductors in detail with necessary diagram and equations. **07**

OR

- Q.4 (a)** Define : **03**
 i. Population Inversion
 ii. Meta stable state
 iii. Active Medium
- (b)** Explain how the materials are classified into conductors, semiconductors and insulators with the help of energy band diagrams. **04**
- (c)** Briefly outline the mechanism of superconductivity using BCS theory and explain penetration depth. **07**
- Q.5 (a)** Compare Direct Band Gap semiconductors and Indirect Band Gap semiconductors **03**
- (b)** What are Excitons? explain the types of Exciton. **04**
- (c)** (i) Estimate the concentration of holes and electrons in n-type silicon at 300 K. If the conductivity is $8 \times 10^4 \text{ } \Omega/\text{m}$. also find these value for p-type Si. $n_i = 2.5 \times 10^{16} \text{ m}^{-3}$, electron and hole mobility is $1600 \times 10^{-4} \text{ m}^2/\text{V}.\text{Sec}$ and $700 \times 10^{-4} \text{ m}^2/\text{V}.\text{Sec}$ respectively. **04**
- (ii) Evaluate the Electron and Hole concentration at room temperature, for a intrinsic semiconductor, if its conductivity at room temperature is $8 \times 10^{-4} \text{ } \Omega/\text{m}$. The mobility of Electron and Hole are $0.24 \text{ m}^2/\text{V}.\text{Sec}$ and $0.08 \text{ m}^2/\text{V}.\text{Sec}$ respectively. **03**

OR

- Q.5 (a)** List difference between drift and diffusion current **03**
- (b)** Explain in detail the Fermi's Golden Rule. **04**
- (c)** (i) In a solid consider an energy level lying 0.2 eV above Fermi level. Estimate the probability of this level not being occupied by an electron at room temperature. **04**
- (ii) A rectangular semiconductor specimen of thickness 1 mm is placed in the magnetic field of flux density 0.5 Wb/m^2 . Current of 1.5 mA is flowing through the specimen in one direction. Evaluate Hall coefficient of the material if generated hall voltage is 7.5 mV. **03**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-I & II EXAMINATION – WINTER 2024****Subject Code:3110018****Date:04-01-2025****Subject Name:Physics****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Define: (1) Valance Band (2) Current density (3) Fermi Function	03
	(b) Distinguish between Direct energy band gap semiconductor and Indirect energy band gap semiconductor with any four points.	04
	(c) Explain in details about Kroning-Penney model and discuss the different cases with necessary diagrams.	07
Q.2	(a) List out any six applications of Solar cell.	03
	(b) The energy gap of two intrinsic semiconductors A and B are 0.36 eV and 0.72 eV respectively. Compare the carrier density of semiconductors A and B at temperature 300 K.	04
	(c) Define density of holes in valence band and derive the necessary equation for it.	07
	OR	
	(c) (1) State and prove the Law of Mass Action.	03
	(2) Define Luminescence and categories them with suitable illustrations.	04
Q.3	(a) Distinguish between Spontaneous Emission and Stimulated emission with any three points.	03
	(b) Define Exciton and classify them in details.	04
	(c) Derive the expression for Fermi energy and carrier concentration in N-type semiconductor.	07
	OR	
Q.3	(a) Distinguish between N-type semiconductor and P-type semiconductor with any four points.	03
	(b) What are the processes known as carrier generation and recombination? Explain the Optical Generation and recombination process for charge carriers.	04
	(c) Derive the relation between Einstein's coefficients and interpret the results.	07
Q.4	(a) Define: (1) Superconductivity (2) Critical Temperature (3) Cooper pair	03
	(b) The values of critical magnetic field for a superconductor are 270 T and 60 T at temperature 20 K and 90 K respectively. Calculate the values of critical temperature and zero magnetic field.	04
	(c) Define density of state and derive the necessary formula for DOS.	07
	OR	
Q.4	(a) Name the types of capacitances are associated with PN junction and briefly explain each of them.	03
	(b) Explain the Hot Point Probe method to verify the type of extrinsic semiconductors.	04
	(c) Explain the Four Probe method and derive the necessary formula to find the resistivity of Bulk material.	07

- Q.5** (a) Distinguish between type-I superconductor and type-II superconductor. **03**
 (b) For a metal having density of electron is 6.5×10^{28} then find the relaxation time of conduction electrons if the metal resistivity is 1.435×10^{-8} ohm-meter **04**
 (c) Explain the Vander Pauw measurement method. Discuss about the measurement of Resistivity of the material by this method. **07**

OR

- Q.5** (a) What is Drude Model? List out the basic assumptions of Drude Model. **03**
 (b) Discuss about Meissner's effect with diagrams and prove that the superconductors are perfect diamagnetic material. **04**
 (c) (1) What is Josephson's Junction? Explain the types of Josephson's effect. **03**
 (2) Calculate the Critical magnetic field and Critical Current for a superconducting wire of Lead having a diameter of 1 mm at 4.2 K. Critical temperature of lead is 7.18 k and zero magnetic field is 65 kA/m. **04**

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER-I&II EXAMINATION – SUMMER 2025

Subject Code:3110018

Date:25-06-2025

Subject Name:Physics

Time:10:30 AM TO 01:00 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

- Q.1** (a) State and discuss Wiedemann-Franz law. **03**
(b) What is called Energy Band Gap? Draw energy band diagram for (i) **04**
Conductors, (ii) Semiconductors and (iii) Insulators.
(c) What is photovoltaic effect? With neat diagram discuss construction and **07**
working of solar cell.

- Q.2** (a) Enlist the assumptions of classical free electron theory. (Any three) **03**
(b) State any four points of differences between N-type and P-type **04**
Semiconductors.
(c) Discuss in detail Kronig – Penney Model. **07**

OR

- (c) Define density of states and derive its expression. **07**

- Q.3** (a) Evaluate Fermi function (or occupation probability) for energy $2k_B T$ above **03**
the Fermi energy.
(b) Derive formula of diffusion current density for electron as well as hole. **04**
(c) Discuss forward bias and reverse bias for a PN junction diode and draw **07**
characteristics curve.

OR

- Q.3** (a) Write short note on: Phonon. **03**
(b) Discuss Fermi golden rule for transition probability. **04**
(c) Derive formula of fermi energy level for N-type semiconductor. **07**

- Q.4** (a) State any three points of differences between spontaneous emission and **03**
stimulated emission.
(b) Discuss direct and indirect band gap semiconductors. **04**
(c) Explain in detail experimental procedure for DLTS. **07**

OR

- Q.4** (a) (i) What is called exciton? **03**
(ii) Name two metal-semiconductor junctions.
(iii) What is called optoelectronic device?
(b) The electrical conductivity (σ) of copper at 300 K is $6.2 \times 10^7 \Omega^{-1} \text{ m}^{-1}$. **04**
Calculate the thermal conductivity of copper. Lorentz number is $2.44 \times 10^{-8} \text{ W/K}^2$.
(c) Discuss in detail UV-VIS method for band gap measurement of **07**
semiconductor.

- Q.5** (a) What is Meissner effect? For superconductor show that $\chi_m = -1$. **03**
 (b) Write any four points of comparison of type-I and type-II superconductor. **04**
 (c) (i) Calculate the critical current for a superconducting wire having radius of 0.2 mm at 5 K. Critical temperature of material of wire is 7.2 K and $H_c(0)$ is 2.2×10^4 A/m. **07**
 (ii) For a superconducting material isotopic mass is 195 amu and critical temperature is 5.2 K. Calculate isotopic mass at 6.5 K.

OR

- Q.5** (a) Write a short note on: SQUID **03**
 (b) What is superconductivity? Discuss any three properties of superconductor. **04**
 (c) Discuss in detail four-point probe method for the measurement of resistivity. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-I & II EXAMINATION – WINTER 2025****Subject Code:3110018****Date:01-01-2026****Subject Name:Physics****Time:02:30 PM TO 05:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed

		Marks
Q.1	(a) Prove that Probability of occupancy of Fermi energy level is 0.5 at $T > 0$ K	03
	(b) Explain how the materials are classified into conductors, semiconductors and insulators with the help of energy band diagrams.	04
	(c) Explain in detail how Kroning and Penney model explains the origin of band gap in semiconductors.	07
Q.2	(a) Define Following Terms (i)Population Inversion , (ii) Meta stable state (iii) Active Medium	03
	(b) Define carrier transport and explore how charge carriers move within semiconductor materials, emphasizing the mechanisms of drift and diffusion.	04
	(c) Describe how to formulate the density of holes in the valence band for intrinsic semiconductors.	07
	OR	
	(c) Describe the definition of carrier concentration in intrinsic semiconductors and show how to derive the formula for electron density in the conduction band.	07
Q.3	(a) Determine the band gap of the material when the light source composed of it emits radiation with a wavelength of 8628 Å.	03
	(b) Explain construction and working of PV solar cell.	04
	(c) Interpret the process of deriving the Fermi level expression for n-Type semiconductors, and subsequently, utilize this expression to compute the electron density in the conduction band.	07
	OR	
Q.3	(a) The intrinsic carrier concentration at room temp in Germanium is $4.17 \times 10^{18} \text{ m}^{-3}$. The mobility of Electron and Hole are $0.29 \text{ m}^2/\text{VSec}$ and $0.16 \text{ m}^2/\text{vSec}$ respectively. Find out its Resistivity	03
	(b) The conductivity of a semiconductor at 30°C is $742 \text{ } \Omega/\text{m}$ and at 90°C is $2100 \text{ } \Omega/\text{m}$. Calculate the Energy Band Gap.	04
	(c) Discuss the Einstein's theory about interaction between matter and radiation in detail. Also discuss the absorption and emission with necessary equation.	07
Q.4	(a) A Silicon plate of thickness 1mm, breadth 10 mm and length 100 mm is placed in a magnetic field of 0.5 Wb/m^2 acting along its thickness. If 0.01 A	03

current flows along its length, calculate the Hall coefficient if developed Hall voltage is 1.83 mV.

- (b) Discuss in detail Four probe method used for resistivity measurements in thin sheet. **04**
- (c) Define density of states and obtain expression for the density of states in terms of mass of electron(m), Planck's constant (h) and Energy (E). **07**

OR

- Q.4**
- (a) What is hot point probe method? Explain **03**
 - (b) List the condition to be satisfied to use Vander Pauw technique and derive expression of bulk resistivity measurement using it. **04**
 - (c) What is Band gap? How can we use UV-VIS spectroscopy for band gap measurement for semiconductors explain in detail. **07**

- Q.5**
- (a) Distinguish between Type-I and Type-II Superconductivity **03**
 - (b) The Fermi level in given material is 2.8 eV. What are the energies for which the probabilities of occupancy at 270K are 0.89, 0.20, 0.02 and 0.5 **04**
 - (c) (i) Discuss BCS theory of superconductivity in brief **07**
(ii) What is cryotron? Discuss

OR

- Q.5**
- (a) Explain the Fermi's Golden Rule in brief. **03**
 - (b) The critical magnetic field of a given superconductor is 9.5×10^3 A/m at 0 K. At what temperature does the critical magnetic field of the sample drop to 5×10^3 A/m, given that the critical temperature of the sample is 7.2 K? Also, calculate the critical current density at that temperature if the radius of the wire is 0.4 mm. **04**
 - (c) List various properties of Superconductors and explain them in detail. **07**
